Backend Configuration & Data Setup – Data Architecture

Overview

The **Backend Configuration & Data Setup** phase establishes the technical foundation of the **Virtual Agent–Driven SLA Breach Awareness & Justification System**. This phase focuses on configuring the core ServiceNow data model, enabling SLA tracking, and preparing the platform for automation and governance.

The **Incident** table serves as the primary source for incident records, while the **Task SLA** table manages SLA execution and monitoring. Supporting components such as **SLA Definitions**, **Business Schedules**, **Update Sets**, and custom incident fields are configured to ensure accurate SLA calculations, audit readiness, and efficient governance.

All configurations are performed using native ServiceNow features without custom scripting, ensuring maintainability and alignment with platform best practices.

Objectives

The objectives of this phase are to:

- Establish the backend architecture for SLA governance.
- Configure data structures required for SLA monitoring.
- Prepare the Incident table with governance-related fields.
- Enable accurate SLA calculations using business schedules.
- Configure Update Sets for version control and deployment.
- Support future automation through a standardized data model.

Data Architecture

The backend architecture consists of the following core components:

1. Incident Table

The **Incident** table is the primary business table used to manage incidents throughout their lifecycle.

Additional custom fields are added to support SLA governance, including:

- SLA At Risk

- SLA Breach Reason
- SLA Breach Justification
- SLA Acknowledged

These fields allow the system to record governance information and support reporting and audits.

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2. Task SLA Table

The **Task SLA** table automatically tracks SLA execution for each incident.

It stores information such as:

- SLA Start Time
- SLA End Time
- Elapsed Time
- Remaining Time
- Percentage Complete
- SLA State

The Task SLA table enables continuous monitoring of SLA progress and identifies incidents approaching breach.

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3. SLA Definitions

Custom SLA Definitions determine:

- SLA duration
- Start conditions
- Stop conditions
- Pause conditions
- Business schedule
- Target response time

These definitions ensure consistent SLA behavior across all incidents.

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4. Business Schedule

A business schedule defines official working days and business hours used for SLA calculations.

Benefits include:

- Accurate SLA timing
- Exclusion of holidays
- Exclusion of weekends
- Consistent business-hour calculations

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5. Update Sets

Update Sets are used to capture every configuration change made during project development.

Benefits include:

- Version control
- Easy migration between instances
- Backup of configurations
- Controlled deployment
- Audit of configuration changes

Backend Configuration Activities

Create Local Update Set

Before starting configuration, create a new Local Update Set to capture all customizations.

Navigation

System Definition → Update Sets → Local Update Sets

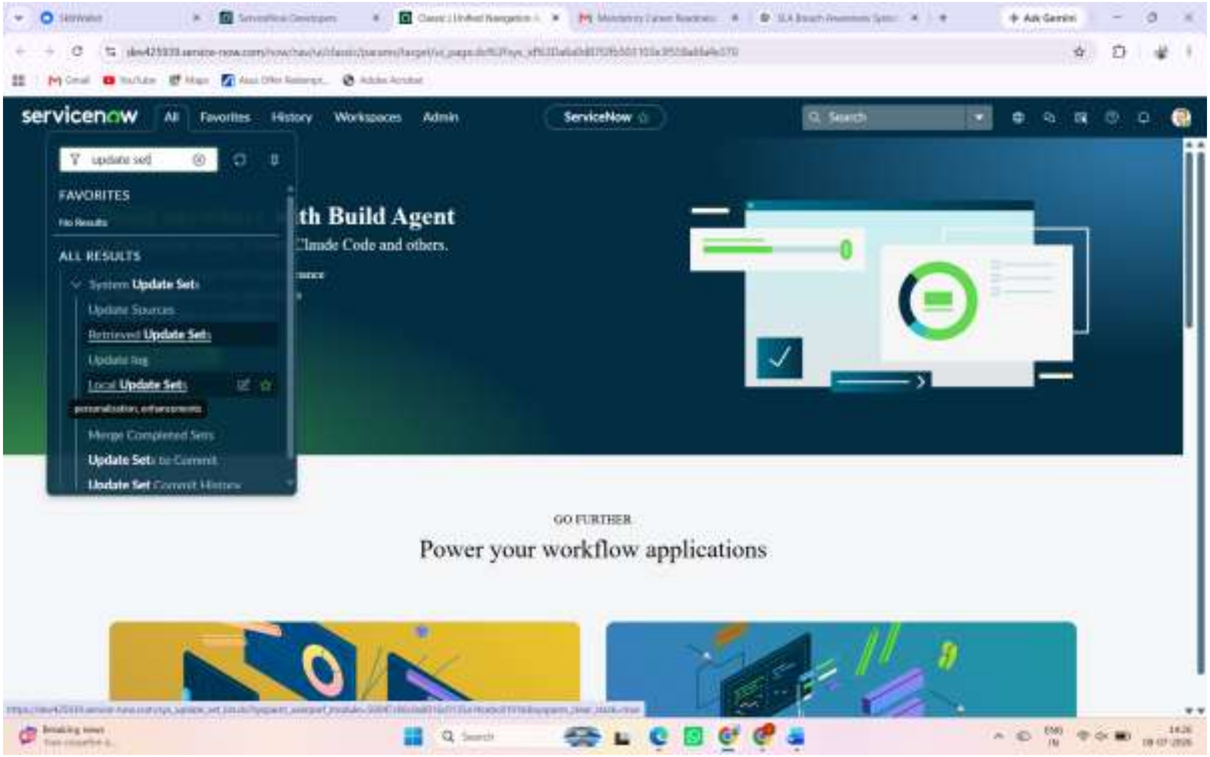
Steps

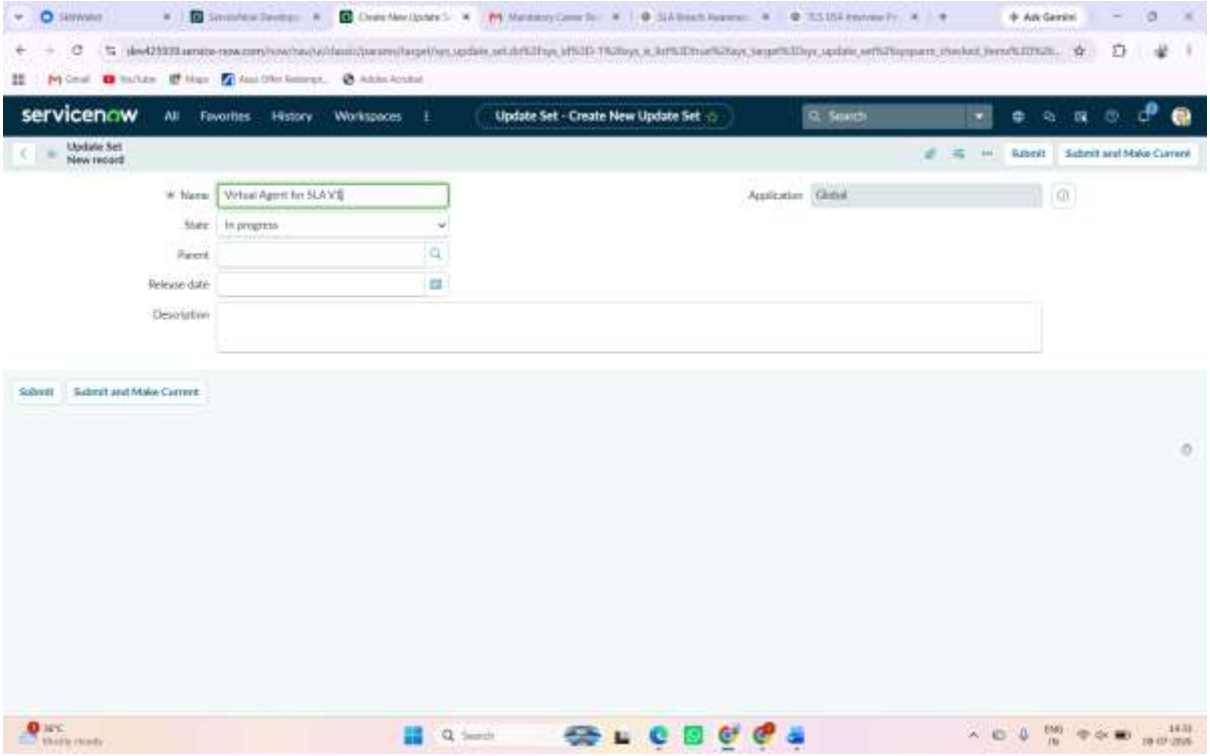
1. Open **Local Update Sets**.
2. Click **New**.
3. Enter the following details:

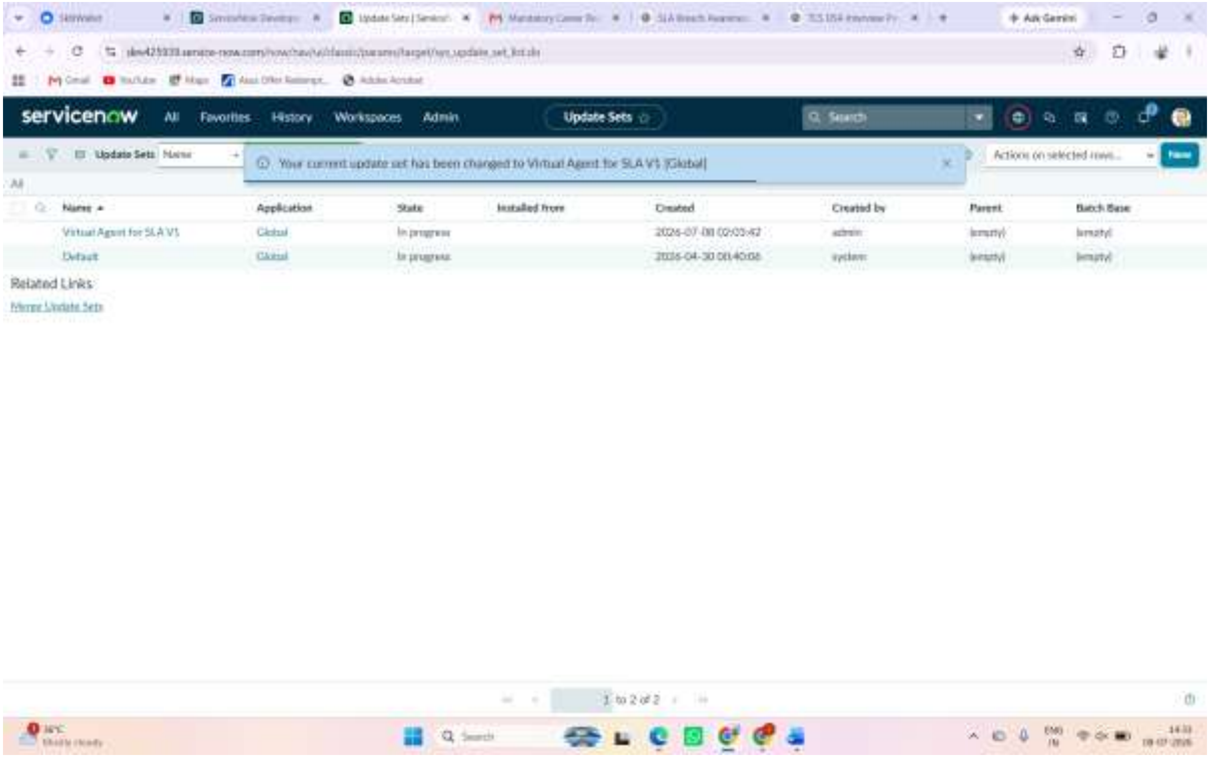
Field	Value
Name	Virtual Agent for SLA V1

4. Click **Submit and Make Current**.
5. Verify that the Update Set status is **Current**.

This ensures that every configuration performed during the project is automatically recorded.







Configure Incident Fields

Create custom fields on the Incident table.

Field Name	Purpose
SLA At Risk	Indicates whether an incident is approaching SLA breach.
SLA Breach Reason	Stores the reason for SLA delay.
SLA Breach Justification	Captures the detailed explanation for the breach.
SLA Acknowledged	Records whether the assignee acknowledged the SLA warning.

Configure Business Schedule

Create a business schedule based on organizational working hours.

Typical configuration includes:

- Monday to Friday
- 9:00 AM – 6:00 PM
- Exclude weekends
- Exclude company holidays

The SLA engine uses this schedule to calculate elapsed and remaining business time accurately.

Configure SLA Definition

Create or configure an Incident Resolution SLA with:

- SLA Name
- Target Duration
- Start Condition
- Stop Condition
- Pause Condition
- Business Schedule
- Applicable Incident Conditions

The SLA automatically attaches to qualifying incidents and begins tracking progress.

Backend Validation

After completing the configuration:

- Verify that the Update Set is active.
- Confirm custom fields appear on the Incident form.
- Ensure the Incident Resolution SLA attaches automatically.
- Validate Task SLA records are generated.
- Confirm the business schedule is applied correctly.
- Check that SLA percentage updates as expected.

Deliverables

- Local Update Set
- Incident Table Configuration
- Custom SLA Governance Fields
- Task SLA Configuration
- SLA Definition
- Business Schedule
- Backend Data Architecture
- Configuration Validation Report

Expected Outcomes

Upon completion of this phase:

- The backend infrastructure is ready for SLA governance.
- Incidents are capable of tracking SLA risk.
- Governance fields are available for future automation.
- SLA calculations follow business working hours.
- All configuration changes are tracked using Update Sets.

- The platform is prepared for Flow Designer, Virtual Agent, UI Policies, reporting, and dashboards.

Conclusion

The **Backend Configuration & Data Setup** phase provides the core infrastructure for the **Virtual Agent–Driven SLA Breach Awareness & Justification System**. By configuring Update Sets, the Incident table, Task SLA records, SLA Definitions, business schedules, and governance fields, this phase establishes a reliable, scalable, and audit-ready backend. The resulting architecture supports accurate SLA monitoring, seamless automation, secure governance, and effective reporting throughout the project lifecycle.